Algebraic Groups

So Murata

WiSe 25/26, University of Bonn

1 Preliminaries

Let F be a non-Archimedian local field (i.e. $F/\mathbb{Q}_p$ for some finite extension of a p-adic numbers or $F = \mathbb{F}_q((t))$ where q is a power of p). We denote $\mathcal{O} \subseteq F$ for the ring of integers.

Example 1.1. *$F = \mathbb{Q}_p, \mathcal{O}_F = \mathbb{Z}_p$, or $\mathbb{F}_q[\![t]\!] \subseteq \mathbb{F}_q((t))$.*

We will denote ϖ for the uniformizer (i.e. a generator of the unique maximal ideal $\mathfrak{p} \subseteq \mathcal{O}$). In one of our cases, we have $\mathbb{F}_q = \mathcal{O}/\varpi\mathcal{O}$.

The topology on $\mathrm{GL}_n(F)$ is defined by the neighborhood basis around the origin. Explicitly, this is constructed by

$$I_n + \varpi \operatorname{Mat}_{n\times n}(\mathcal{O}) \supseteq I_n + \varpi^2 \operatorname{Mat}_{n\times n}(\mathcal{O}) \supseteq \cdots$$

Definition 1.1. *Let $G = \mathrm{GL}_n(F)$ be a group. A smooth representation of G is a pair (π, V) such that*

1. *V is a complex vector space.*
2. *$\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ is a group homomorphism.*
3. *$\forall v \in V$, $\mathrm{Stab}_G(v) := \{g \in G \mid \pi(g)(v) = v\}$ is open.*

Example 1.2. *For $\mathrm{GL}_1(\mathbb{Q}_p) = \mathbb{Q}_p^\times$, $V = \mathbb{C}$ the we can take,*

$$\pi(g) = \mathrm{ab}\, g_p = p^{-\mathrm{val}(g)} \in \mathrm{Aut}(\mathbb{C}) \cong \mathbb{C}^\times.$$

Definition 1.2. *A representation (π, V) is irreducible if (π, V) has precisely two sub-representations (i.e. the trivial representations).*

Remark 1.1. *By the definition above, $\{0\}$ is not an irreducible representation.*

Remark 1.2. *All irreducible smooth representations of $\mathbb{Q}_p^\times$ are one dimensional.*

1.1 Linear Algebraic Groups

Definition 1.3. *Let k be an algebraically closed field. A linear algebraic group over k is a reduced Zariski-closed subgroup of GL_n over k for some $n \in \mathbb{N}$.*

Equivalently, linear algebraic groups over k are smooth affine group schemes of finite type over k.

Notation 1.1. *From now on, $\mathbb{G}$ will denote the linear algebraic groups and $G = \mathbb{G}(k)$ the k-points of $\mathbb{G}$.*

Example 1.3. $\mathrm{GL}_n(k), \mathrm{SL}_n(k), \mathrm{O}_n(k), \mathrm{Sp}_{2n}(k)$ *are linear algebraic groups and,* $B(k) \subseteq \mathrm{GL}_n(k)$*, upper triangular matrices*

Definition 1.4. $G \subseteq \mathrm{GL}_n(k)$ *is unipotent if G is conjugate to a subgroup of upper triangular matrices with* 1*-s in the diagonal.*

Definition 1.5. *The unipotent radical (denoted by $(RG)_u$ or R_uG) of a linear algebraic group is the maximal connected unipotent normal subgroup of G.*

Definition 1.6. *An algebraic group G is reductive if its unipotent radical is trivial.*

Example 1.4. $\mathrm{GL}_n(k), \mathrm{SL}_n(k), \mathrm{SO}_n(k), \mathrm{Sp}_{2n}(k)$ *are all reductive.* $U_n(k)$ *and* $B(k)$ *are not.*

Definition 1.7. *A linear algebraic group G is a torus if $G \cong (k^\times)^n$ for some $n \in \mathbb{N}$.*

Theorem 1.1. *All maximal tori in an linear algebraic group G are conjugate.*

Notation 1.2. *T will be a maximal torus in G. Then we will denote $\mathfrak{g} = \mathrm{Lie}(G) := T_1(G)$ with Lie algebra structure.*

Example 1.5.

Linear Algebraic Groups	*Lie Algebra*	*Commutator*
GL_n	$\mathrm{Mat}_{n\times n}$	$[A, B] = AB - BA$
SL_n	$\{A \in \mathrm{Mat}_{n\times n} \mid \mathrm{tr}(A) = 0\}$	$[A, B] = AB - BA$
Sp_{2n}	$\left\{ A \in \mathrm{Mat}_{n\times n} \;\middle|\; \mathrm{tr}(A) = 0, JA + {}^tAJ = 0 \right\}$	$[A, B] = AB - BA$

Remark 1.3. *G acts on $\mathfrak{g}$ via the adjoint action. That is the derivation of the conjugation action on itself. Explicitly, for $g \in \mathrm{GL}_n(k)$ and $A \in \mathrm{Mat}_{n\times n}(k)$, we define,*

$$\mathrm{Ad}(g)(A) = gAg^{-1}.$$

Hence T also acts on $\mathfrak{g}$ and such induced actions are diagonalizable.

Example 1.6.

$$\mathrm{diag}(t_1, \cdots, t_n)E_{ij} = t_i t_j^{-1} E_{ij}.$$

Definition 1.8. $X^*(T) = \mathrm{Hom}_k(T, \mathbb{G}_m)$ *the characters of T.*

Example 1.7. $X^*(\mathbb{G}_m(k)) = \mathbb{Z}$ *explicitly, this sends $k^\times \ni x \mapsto x^n \in k^\times$ for some n.*

From now on, G will denote the connected reductive group.

Proposition 1.1. *For $\mathfrak{g} = \mathrm{Lie}(G)$. We have,*

$$\mathfrak{g} = \bigoplus_{\alpha \in X^*(T)} \mathfrak{g}_\alpha,$$

where $\mathfrak{g}_\alpha$ is the subspace such that T acts via the character α. Explicitly, this is

$$\mathfrak{g}_\alpha := \{X \in \mathfrak{g} \mid \forall t \in T, \mathrm{Ad}(t)(X) = \alpha(t)X\}.$$

The root of G is

$$\Phi(G) = \Phi(G; T) := \{\alpha \in X^*(T) \backslash \{0\} \mid \dim \mathfrak{g}_\alpha \neq 0\}.$$

Remark 1.4. $\dim \mathfrak{g}_\alpha = 1$ *for all* $\alpha \in \Phi(G)$ *and* $\Phi(G) \subseteq X^*(T) \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}^n$ *which is a root system. Such root systems are classified via Dynkin diagrams.*

The root system $\Phi(G) = \Phi(G;T)$ *is independent of* T *up to isomorphisms.*

$G/Z(G)$ *is uniquely determined by* $\Phi(G)$*.*

Example 1.8. *For* $G = \mathrm{GL}_n(k)$ *and* $\mathfrak{g} = \mathfrak{gl}_n$*, we have,*

$$\mathrm{Mat}_{n\times n}(k) = \bigoplus_{\alpha \in X^*(T)} \mathfrak{g}_\alpha.$$

This is

$$T \oplus \bigoplus_{i\neq j} \langle E_{ij} \rangle.$$

Hence,

$$\Phi(\mathrm{GL}_n) = \{\mathrm{diag}(t_1, \cdots, t_n) \mapsto t_i t_j^{-1} \mid 1 \leq i \neq j \leq n\}.$$

We have

$$X^*(T) = \mathbb{Z}t_1 \oplus \mathbb{Z}t_2.$$

The root system is



Also do for GL_3

Remark 1.5. *A reductive group is called semisimple if its center is finite. Root system does not determine the groups however, they do determine groups modulo their center.*

References

[1] Borel, Linear algebraic groups

[2] Humphreys, Linear algebraic groups

[3] Springer, Linear algebraic groups

[4] Conrad, Linear algebraic groups.

[5] Vignéras, Marie-France Représentations l-modulaires d'un groupe réductif p-adique avec $l \neq p$

[6] Roche, Alan, The Bernstein Decomposition and the Bernstein Centre.

[7] Debacker, Stefan, Lecture notes https://websites.umich.edu/ smd-backr/MATH/notes.pdf.

[8] Renard, David (2010), Representations des groupes reductifs p-adiques Séminaire Bourbaki page 193-219, Société Mathématique de France.

[9] Buschnell, Colin, Kutzko, Phillip (1998), Smooth representations of reductive p-adic groups: structure theory via types, Proceedings of the London Mathematical Society.

[10] Moy, A., Prasad, G. Jacquet functors and unrefined minimal K-types. Commentarii Mathematici Helvetici 71, 98–121 (1996).

[11] Fintzen, Jessica. (2020), Types for tame p-adic groups. Annals of Mathematics.

[12] Ohara, Kazuma. A comparison of endomorphism algebras.

[13] Blondel, Corinne. Quelques propríetés des paires couvrantes.

[14] Dat, Jean-Francois. Finitude pour les representations lisses de groupes p-adiques.

[15] Solleveld, Maarten. Affine Hecke algebras and their representations.

[16] Jeffrey D. Adler, Jessica Fintzen, Manish Mishra, Kazuma Ohara. Structure of Hecke algebras arising from types.

[17] Jeffrey D. Adler, Jessica Fintzen, Manish Mishra, Kazuma Ohara. Reduction to depth zero for tame p-adic groups via Hecke algebra isomorphisms.